Chemistry 3A – Fall 2025 Sections 43957-43958 **Version B**

Midterm Examination #4

Select the **BEST** response for the question. Point values: multiple choice 3 pt; true/false 1 pt

1. Which statement accurately defines the **limiting reactant** in a chemical reaction?
   a) The reactant with the largest molar mass used in the reaction
   b) The reactant remaining in the largest amount after the reaction is complete
   c) The reactant that determines the maximum amount of product that can be formed
   d) The reactant with the smallest coefficient in the balanced chemical equation

2. In gas stoichiometry calculations, what is the molar (i.e., volume of 1 mole) of any ideal gas at Standard Temperature and Pressure (STP)?
   a) 1 L b) 22.4 L c) 8.314 L atm/mol K d) 6.022×10^{23} L

3. For the equilibrium reaction $H_2 (g) + Cl_2 (g) \rightleftharpoons 2 HCl (g)$, and $[H_2] = 0.0001$ M, $[Cl_2] = 0.0001$ M, and $[HCl] = 2500$ M, which of the values below is the value of K?
   a) 2.50×10^{15} b) 6.25×10^{14} c) 2.50×10^{8} d) 6.25×10^{16}

4. Which of the following is a general property of an acid?
   a) Feels slippery to the touch b) Turns red litmus paper blue
   c) Reacts with active metals to produce hydrogen gas, H_2
   d) Has a pH value greater than 7.00 in aqueous solution

5. Which of the following pairs are conjugates in acid-base chemistry?
   a) HCl / Cl^- b) NH_3 / NH_2^- c) H_2O / OH^-
   d) H_2SO_4 / HSO_4^- e) All of (a), (b), (c) and (d)

6. Which statement explains why water is amphoteric, that is, acts as an acid or a base?
   a) It can donate a proton to form OH^-
   b) It can accept a proton to form H_3O^+
   c) It can act as both acid and base depending on the reaction partner.
   d) (a), (b), and (c) are all true in explaining it

7. Of the statements below, select the best that is true of ${}^{27}_{13}\mathrm{Al}$?
   a) It has 40 neutrons and protons b) It has 13 protons
   c) It has 27 neutrons d) all the above e) none of the above

8. A solution has $[H^+] = 1.0 \times 10^{-5}$ M. What is the pH?
   a) 1.00 b) 5.00 c) 9.00 d) 14.00

9. Which of the following is a strong base?
   a) KOH b) NH_3 c) HCl d) NaCl e) CO_2

10. 10.00 mL of acetic acid solution was titrated with 0.050 M NaOH solution with the initial buret reading at 2.75 mL and the final buret reading at 22.00 mL. What is the acetic acid concentration?
    a) 0.019 M b) 0.096 M c) 0.124 M d) 0.048 M

11. In a thermochemical equation, the energy value written on the product side indicates:
    a) The reaction is endothermic b) The reaction requires a catalyst
    c) The reaction is exothermic d) The reaction occurs at constant volume.

12. If a weak acid has an equilibrium constant K_a = 5.0 x 10^{-2}, then what is the equilibrium constant K_b of its conjugate base? Note K_w = 1.0 x 10^{-14}
  a) 2.0 x 10^{-16} b) 5.0 x 10^{-13} c) 2.0 x 10^{-13} d) 2.0 x 10^{-10}

13. Consider the following equilibrium: PCl_3 (g) + Cl_2 (g) ⇌ PCl_5 (g). If large amount of PCl_5 product is added to container, in which direction will equilibrium shift to restore balance?
  a) Shift to the right (forward reaction) to produce more PCl_5.
  b) Shift to the left (reverse reaction) to consume the added PCl_5.
  c) There will be no shift, as adding a product does not affect the equilibrium
  d) The equilibrium constant *K* will increase

14. What is a Geiger-Müller detector?
  a) It is used to capture neutrons to control fission chain reactions
  b) It is used to count radioactive decay events
  c) It is used to determine product and reactant concentrations to calculate equilibrium constants K
  d) It is another kind of calorimeter used to calculate changes in enthalpies of reaction

15. What is the radioisotope of an element that is a gas that can accumulate in homes and is responsible particularly for causing nonsmoking-related lung cancers?
  a) Xenon-133 b) Krypton-85 c) Radon-222 d) Carbon-14

16. What is the purpose of a buffer?
  a) It is a weak acid or a weak base
  b) when present in a solution, it will resist changes in pH caused by addition of strong acids or bases
  c) It is used as shielding for all radiation, alpha and beta particles and gamma rays
  d) both a) and b)

17. For AgCl (s) ⇌ Ag^+ (aq) + Cl^- (aq), if [Ag^+] = 1.0 × 10^{-1} M and [Cl^-] = 1.0 × 10^{-1} M, what is the calculated value of K_{sp}?
  a) 1.0 × 10^{-1} b) 1.0 × 10^{-2} c) 2.0 × 10^{-1} d) 2.0 × 10^{-2}

18. What are the features of a catalyst?
  a) It increases the activation energy (E_a) in the reaction progress curve
  b) it is never consumed in the reaction
  c) It stops a chemical reaction immediately
  d) it stops the equilibrium constant *K* from being temperature-dependent

19. Suppose a reaction with gases X, Y, and Z in which the reaction is **3** X (g) + Y (g) ⇌ **2** Z (g). Note the coefficients. What will happen if the pressure is **increased** on the system according to Le Chatelier's Principle?
  a) The reaction will favor the formation of the reactants (the reverse direction)
  b) The reaction will favor the formation of the products (the forward direction)
  c) The reaction will change the reaction to **3** X (g) + **2** Y (g) ⇌ **2** Z (g)
  d) There will be no effect: the amounts of reactants and product remain the same

20. Initially you have 100.0 g of U-235 radioisotope. How much do you have left if three (3) half-lives pass?
  a) 100.0 g b) 50.00 g c) 25.00 g d) 12.50 g

21. What do the brackets '[ ]' mean in this notation: $[SO_4^{2-}]$
   a) it indicates the molarity of sulfate ion
   b) it indicates the moles per liter of sulfate ion
   c) it is used to indicate the concentration of sulfate ion in solution
   d) all of the above
   e) none of the above

22. What is true about the concept of chemical equilibrium?
   a) chemical reactions only go one direction and have only a right-facing arrow
   b) chemical reactions can be reversible
   c) it only applies to what happens when the reactants and products are gases
   d) it states that reactants completely disappear in a chemical reaction

23. Which of the choices below shows the correct **mass** of iron(III) oxide product made if 223.4 g of iron metal reactant is used in the reaction $4\ Fe\ (s) + 3\ O_2\ (g) \rightarrow 2\ Fe_2O_3$. Molar mass of Fe = 55.85 g/mol and Fe_2O_3 = 159.69 g/mol
   a) 159.69 g b) 319.38 g c) 4.00 mol d) 572.68 g

24. According to the Bronsted-Lowry definition of acids and bases, which statement is correct?
   a) an acid is a proton acceptor
   b) an acid is a proton donor
   c) a base is a proton donor
   d) a base is an electron pair acceptor

25. Which of the following factors does not affect a chemical reaction rate?
   a) Color of reactants
   b) temperature
   c) concentration
   d) orientation of collisions

Continue to next page for True-False questions

26. The half-life of a radioisotope is the time required for half of the sample to decay.
   a) true   b) false
27. In performing a chemical reaction, the theoretical yield is the maximum amount of product that can be formed from the initial amounts of reactants, while the actual yield is what you actually got from doing the reaction
   a) true   b) false
28. **Alpha particles** are the most penetrating type of common radiation and require thick concrete or lead shielding
   a) true   b) false
29. A strong acid differs from a weak acid primarily because the strong acid is present at a higher concentration
   a) true   b) false
30. Balancing a chemical reaction equation is done by changing the subscripts on the atoms of one or more of the compounds in the reaction equation
   a) true   b) false
31. Collision theory in part states that for a chemical reaction to occur, particles must collide with correct orientation and sufficient energy
   a) true   b) false
32. A catalyst lowers the activation energy (E_a) of a reaction but does not change ΔH.
   a) true   b) false
33. If the temperature of an endothermic reaction at equilibrium is raised, Le Chatelier's Principle predicts the equilibrium will shift to favor the reverse reaction (exothermic direction)
   a) true   b) false
34. Water (H_2O) can act as both an acid and a base depending on the reaction partner
   a) true   b) false
35. If the pH of a solution is 6.00, the corresponding pOH of the solution must be 10.00 because pH + pOH = 16.00
   a) true   b) false

Scratch